INCH-POUND

MIL-DTL-46063H
14 September 1998
SUPERSEDING
MIL-A-46063G
30 December 1992

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY, 7039

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 7039 wrought aluminum alloy armor plate. The nominal thickness of armor plate covered by this specification is 1/2 inch to 4 inches, inclusive (see 3.7 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V_{50} Ballistic Test For Armor

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Research Laboratory, Weapons and Materials Research Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9535

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.2 - Dimensional Tolerances for Aluminum Mill Products (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B557	Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products (DoD adopted)
ASTM E34	Chemical Analysis of Aluminum and Aluminum-Base Alloys (DoD adopted)
ASTM E227	Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique (DoD adopted)
ASTM E607	Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique, Nitrogen Atmosphere (DoD adopted)
ASTM E716	Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251	Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Argon Atmosphere, Point-to-Plane, Unipolar Self-Initiating Capacitor Discharge (DoD adopted)
ASTM G38	Making and Using C-Ring Stress-Corrosion Test Specimens (DoD adopted)
ASTM G47	Determining Susceptibility to Stress-Corrosion Cracking of High Strength Aluminum Alloy Products (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be manufactured by the process proposed for use on production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

3.2 Chemical composition. Chemical composition shall be within the limits shown in table I. A certification of the chemical composition of the alloy shall be furnished with the ballistic test plates (see 6.2).

TABLE I. Chemical composition.

Element	Percent
Zinc	3.5 - 4.5
Magnesium	2.3 - 3.3
Manganese	0.10 - 0.40
Copper	.10 max
Iron	.40 max
Silicon	.30 max
Chromium	.15 - 0.25
Titanium	.10 max
Others, each	.05 max
Others, total	.15 max
Aluminum (by difference)	Remainder

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of the test specimen taken in the long-transverse direction shall meet the requirements of table II when tested as specified in 4.7.2. If mechanical properties differ from the values shown in table II, or if any other properties are required and if the difference in properties are negotiated between the procuring activity and the supplier, the minimum acceptable ballistic requirements of appendix A will apply.

TABLE II. Mechanical properties.

Thickness, inch	Tensile strength min, psi	Yield strength 0.2% offset, min, psi	Elongation min, percent in 2 in.
Up to 1.500, incl	60,000	51,000	9
Over 1.500	57,000	48,000	8

3.4 Stress corrosion resistance. Unless otherwise specified in the contract or order (see 6.2), plate material 0.72 inch and thicker in the short transverse direction shall be resistant to stress corrosion cracking. A minimum of 5 of the 9 specimens tested shall show no evidence of cracking at the end of 96 hours when tested as specified in 4.7.3 (see 6.4).

3.5 Heat treatment. Heat treatment shall be such as to meet the requirements of this specification (see 6.5).

3.6 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in appendix A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

3.7 Dimensions. Dimensions shall be as specified in the contract or order (see 6.2).

3.7.1 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates submitted for acceptance shall not vary from the specified dimensions by an amount greater than the tolerances shown in ANSI H35.2. Ballistic test plate tolerances shall be ± l/2 inch.

3.7.1.1 Thickness. Thickness tolerances for production armor and ballistic test plates shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered Thickness, inches	Tolerance, inch Minus	Plus 1/
0.500 to 0.625, incl	0.025	
0.626 to 0.875, incl	.030	
0.876 to 1.125, incl	.035	
1.126 to 1.375, incl	.040	
1.376 to 1.625, incl	.045	
1.626 to 1.875, incl	.052	
1.876 to 2.250, incl	.060	
2.251 to 2.750, incl	.075	
2.751 to 3.000, incl	.090	
3.001 to 4.000, incl	.110	

1/ The values for the plus column are derived from the full range of tolerances specified (for sheet and plate not specified for aerospace applications) in table 3.1 of ANSI H35.2 less the value shown in the minus column. For example, the plus tolerance on a 2.0 inch thick by a 73 inches wide plate would be as follows:

Thickness tolerance (from ANSI H35.2)	± 0.060"
Tolerance value shown in minus column	0.060"
Full tolerance less minus tolerance	0.120" - 0.060" = 0.060"
Derived plus tolerance	0.000"

3.7.1.2 Ballistic test plate tolerance. Tolerances shall be ± 1/2 inch on the length and width.

3.8 Marking for identification. Each plate shall be marked with the manufacturer's name or trade mark, number of this specification, the plate thickness in inches, the alloy designation, and the lot number or code relating to the lot number. The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used, unless permitted by the procuring activity (see 6.2).

3.8.1 Ballistic test plates. In addition to the marking in 3.8 each ballistic test plate shall be marked with the letters PRE for first article test plates and ACC for acceptance test plates. This marking shall be impression stamped on the rolled surface within a 1/2-inch band at the edge of the test plate in letters not less than 3/8 inch high. Ballistic retest plates shall be marked "R1" and R2", respectively (see A.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

3.8.2 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

3.9 Ballistic test plate information. A properly executed check list for armor data shall be submitted with each ballistic test plate containing the information as specified in MIL-STD-662.

3.10 Workmanship. Plate shall be uniform in quality and clean, smooth, and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations, and other defects which may affect its use.

3.10.1 Surface and edge condition. Surface cracks, edge cracks, or edge laminations shall be cause for rejection.

4. VERIFICATION

4.1 Clarification of inspections. The inspection requirements specified herein are classified as conformance inspection (see 4.2).

4.2 Conformance inspection. Test for acceptance of individual lots consist of testing all requirements specified in section 3. Verification conformance inspection is in accordance with the provisions herein. Ballistic tests shall be performed at a Government test activity selected by the procuring activity (see 6.2).

4.3 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3.1).
b. Acceptance of individual production lots (see 4.3.2).

4.3.1 First article inspection. First article inspection shall consist of the following:

a. Chemical analysis (see 3.2, 4.5.1.1 and 4.7.1).
b. Tension tests (see 3.3, 4.5.1.2 and 4.7.2).
c. Stress corrosion test (see 3.4, 4.5.1.3 and 4.7.3).
d. Ballistic tests (see 3.6, 4.5.1.4, 4.7.4 and appendix A).
e. Dimensions (see 3.7 and 4.6.2).
f. Identification marking (see 3.8 and 4.6.1).
g. Workmanship (see 3.10 and 4.6.1).

4.3.2 Production lot acceptance inspection. Lot acceptance inspection shall consist of the following:

a. Chemical analysis (see 3.2, 4.5.2.1, 4.5.2.2 and 4.7.1).
b. Tension tests (see 3.3, 4.5.2.3 and 4.7.2).
c. Stress corrosion test (see 3.4, 4.5.2.4 and 4.7.3).
d. Ballistic tests (see 3.6, 4.5.2.5, 4.7.4 and appendix A).
e. Dimensions (see 3.7 and 4.6.2).
f. Identification marking (see 3.8 and 4.6.1).
g. Workmanship (see 3.10 and 4.6.1).
h. Preparation for shipment (see 3.8.2 and section 5).

4.4 Lot. A lot shall consist of plates of the same ordered thickness which have been processed together to produce uniform properties from ingots whose chemical composition was certified as meeting the requirements of this specification. The weight of the finished plates in the lot shall not exceed 50,000 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

4.5 Sampling.

4.5.1 For first article testing.

4.5.1.1 Chemical composition. Samples for chemical analysis shall be removed from the plate material being selected for the ballistic tests.

4.5.1.2 Mechanical properties. One tension test sample shall be removed from the same plate material that has been selected for the ballistic test.

4.5.1.3 Stress corrosion tests. The stress corrosion test samples shall be removed from the same plate that has been selected for the ballistic test.

4.5.1.4 Ballistic tests. Two plates 12 inches by 36 inches (+ 1, -0 inches) from the ordered thickness shall be submitted for ballistic testing. The orientation of these plates with respect to the rolling direction shall be at the option of the manufacturer. The ballistic test may be waived at the discretion of the procuring activity if the manufacturer within 18 months has produced acceptable plate to within ± 0.250 inch of the ordered thickness, provided also that test conditions are the same as for previously accepted plates.

4.5.2 For acceptance of production lots.

4.5.2.1 Chemical composition, ingot analysis. At least one sample shall be taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the procuring activity at the producer's facility.

4.5.2.2 Chemical composition, finished product analysis. When sampling has not been made in accordance with 4.5.2.1, one sample shall be taken for each 4,000 pounds or less in each lot in accordance with ASTM E34 or ASTM E716. Complete product analysis records shall be available to the procuring activity at the producer's facility.

4.5.2.3 Mechanical properties. From each lot, samples for tension tests shall be selected in accordance with table IV. Each sample shall be selected from a different plate in the lot. Should a lot consist of only one plate, only one tension test sample shall be taken (see 6.2).

TABLE IV. Number of tension tests.

Lot size, pounds	Number of samples
To 8,000, incl	2
8,001 to 12,000, incl	3
12,001 to 20,000, incl	4
20,001 up	5

4.5.2.4 Stress corrosion tests. From each lot, one plate 12 inches by 12 inches (± 1 inch), by the ordered thickness, shall be selected for stress corrosion tests. At the discretion of the procuring activity, reduced sampling may be instituted (not every lot need be sampled) (see 4.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

4.5.2.5 Ballistic testing. One plate, 12 inches by 36 inches (+1, -0 inches) from the ordered thickness representing the lot, shall be selected for ballistic testing. The orientation of these plates with respect to the rolling direction shall be at the option of the manufacturer (see 6.2).

4.6 Examination.

4.6.1 Visual. Each plate shall be examined to determine compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.6.2 Dimensional. Plates within a lot shall be measured to determine compliance with the requirements for dimensions and tolerances (see 3.7) in accordance with the sampling procedures approved by the procuring activity.

4.7 Tests.

4.7.1 Chemical composition. Test samples shall be prepared and tested in accordance with ASTM methods E34, E227, E607, or E1251. In case of dispute, the analysis by method ASTM E34 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Test specimens shall be taken in the long transverse direction.

4.7.3 Stress corrosion. Short transverse stress corrosion test specimens shall be prepared and tested in accordance with the procedure outlined in ASTM G38 and G47. The specimens shall be stressed 35 ksi. Nine specimens shall be tested per sample per lot. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.7.3.1 Report of results. The report shall include the following:

a. The producer and lot number.
b. Specification and contract number.
c. "Stress-corrosion test passed" or "stress-corrosion test failed" in accordance with results of the test.

4.7.4 Ballistic testing. Ballistic testing shall be in accordance with appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and appendix A to determine the test projectile and the required V_{50} protection ballistic limit respectively. A minimum of four thickness measurements, at least 2 inches from any edge, are to be taken at random locations in the area to be impacted. A calibrated deep throat micrometer or a calibrated ultrasonic measuring device, is to be used and the individual measurements are to be read to the nearest 0.001-inch, will be reported as the actual thickness of the plate. This thickness will be used to determine the minimum ballistic limit, BL(P), requirement from the appropriate table in the appendix. When necessary, interpolation between two consecutive thicknesses in the table will be performed to determine the minimum required BL(P).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

TABLE V. Acceptance ballistic test plates.

Ordered thickness, inches	Projectile	Angle of Obliquity in degrees
0.500 - 0.749	Cal. .30 APM2	30
0.750 - 0.950	Cal. .50 FSP	0
0.951 - 1.500	Cal. .30 APM2	0
0.951 - 1.500	20mm FSP	0
1.501 - 3.000	Cal. .50 APM2	0
3.001 - 4.000	14.5mm API, BS41	0

4.8 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, and 4.8.2.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3.1, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails the lot shall be permanently rejected with no further testing permitted.

4.8.2 Stress corrosion. If the manufacturer so desires, a retest shall be performed using 13 specimens. A minimum of seven specimens shall be uncracked upon examination at the end of 96 hours of exposure. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2., 4.7.3 and 4.7.4.

4.8.3 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.9 Reduced testing. At the discretion of the procuring activity, (see 6.2) the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirement is being produced, and providing the manufacturer agrees to maintain the same manufacturing procedures. Reduced testing shall be in accordance with sampling plans approved by the procuring activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The aluminum alloy armor covered by this specification is military unique. This material has minimum ballistic limit protection requirements as specified in tables A-I through A-VI. The material will be used in military vehicles to protect the operator and passengers against small arms fire and shrapnel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Nominal thickness of aluminum armor plate (see 1.1).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.3 and A.2.1.1).
d. When first article is required (see 3.1).
e. Mechanical properties and chemical composition of aluminum armor (see 3.2 and 3.3).
f. When stress corrosion testing is required (see 3.4).
g. Drawings, dimensions, and tolerances (see 3.7 and 3.7.1).
h. Marking for identification (see 3.8).
i. Destination of test samples (see 4.2).
j. Locations of tension test specimens (see 4.5.2.3).
k. Locations for determination of ballistic limits (see 4.5.2.5).
l. Rejection and retest requirement, if other than in 4.8.
m. When reduced testing is approved (see 4.9).
n. Packaging requirements (see 5.1).

6.2.1 Armor material test reports. When this specification or specification requirements are used in an acquisition, and data are required to be delivered, the data requirements identified in 3.9 should be developed and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. The data specified should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3 Chemical analysis. ASTM methods that can be used for determination of chemical composition are listed in 4.7.1.

6.4 Stress corrosion. It should be noted that a structure built from plate meeting the requirements prescribed herein should have susceptibility to stress corrosion minimized by proper attention to design and fabrication characteristics. The stress corrosion susceptibility of plate meeting the requirements of this specification will represent a hazard, especially in the short transverse direction. Exposed short transverse sections should be considered carefully in design and protected by special fabrication practices to guard against cracking.

6.5 Mechanical properties to ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure aluminum armor plate meeting the specified ballistic requirements (see 3.6).

6.6 Stress corrosion test. The manufacturer may, at his own risk, ship material prior to the completion of the stress corrosion test.

6.7 New contracts sponsored by government agencies. At the time that a new contract is initiated for the production of combat vehicles, the contractor's supplier is to estimate for the contractor the number, size and delivery schedule of the ballistic test plates which are to be submitted for first article acceptance testing. A lead time of 60 days after the contract has been signed is

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

to be allowed prior to shipment of the first ballistic test plate(s) to APG to insure that all administrative functions for the establishment of a new project have been completed in preparation for the test. The contracting government agency is to initiate the new project through a letter to Commander, US Army Test and Evaluation Command, ATTN: AMSTE-TM, Aberdeen Proving Ground, MD 21005-5005 requesting a cost estimate for the ballistic testing of the applicable number and sizes of plates. In the case of increases in scope of existing projects, similar correspondence is needed.

6.8 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

To go from English	Multiply by	To get Metric SI unit
inch	0.0254	metre (m)
foot	0.3048	metre (m)
foot-lb	1.3558	joule (J)
feet/sec	0.3048	metre per second (m/s)
pounds/sq. inch	0.00689	mega pascal (mpa)

NOTE: Conversion factors can be obtained from ASTM E380 entitled "Metric Practice Guide".

6.9 Definitions.

6.9.1 Contractor. The term "contractor" is defined as the organization having a direct contract with the procuring activity.

6.9.2 Manufacturer. The term “manufacturer” is defined as the organization actually performing the operations covered by this specification.

6.9.3 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.10 Subject term (key word) listing.

Ballistic limit
Ballistic testing
Chemical analysis
Mechanical properties
Obliquity
Projectile

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

ARMOR PLATE, ALUMINUM ALLOY, 7039

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, 7039 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V_{50} Ballistic Test For Armor

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact should be considered fair when an unyawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration should be considered a partial penetration.

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required ballistic limit shall be in accordance with the values shown in tables A-I through A-VI.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V_{50} Ballistic Test For Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature conditioning. Prior to the test, the test item(s) will be temperature conditioned at least eight hours. Thermostatic control will be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) will be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the (PP) velocity being higher than the low (CP) velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit will be computed on the basis of 4 or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing will continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits will be reported as agreed upon between contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, will be fired to reduce the gap to 25 fps or less. The ballistic limit will then be recomputed using the above criteria. The recomputed BL(P) will be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

A.5.2.2 First article (retests). Submission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests must pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision will depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his own expense two additional test plates from the same lot for ballistic retest. If either of these plates fail the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fail, the lot shall be permanently rejected.

A.5.3 Disposition of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates will be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates will be returned, upon request, as in A.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps		Thickness, in.	Required BL(P), fps
	0.475	1378	0.580	1604		0.685	1804
	.480	1390	.585	1614		.690	1813
	.485	1401	.590	1624		.695	1822
	.490	1413	.595	1634		.700	1831
	.495	1424	.600	1644		.705	1839
*	.500	1435	.605	1654		.710	1848
	.505	1446	.610	1664		.715	1857
	.510	1457	.615	1674		.720	1866
	.515	1468	.620	1683		.725	1874
	.520	1479	.625	1693		.730	1883
	.525	1490	.630	1702		.735	1891
	.530	1501	.635	1712		.740	1900
	.535	1511	.640	1721		.745	1909
	.540	1522	.645	1730	**	.749	1915
	.545	1533	.650	1739		.750	1917
	.550	1543	.655	1748		.755	1926
	.555	1554	.660	1758		.760	1934
	.560	1564	.665	1767		.765	1942
	.565	1574	.670	1776		.770	1950
	.570	1584	.675	1786		.775	1959
	.575	1594	.680	1795		-	-

* Specification requirements begin for this ordered thickness.

** Specification requirements end for this ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
0.700	1666	0.790	2012	0.880	2427
.705	1683	.795	2033	.885	2452
.710	1701	.800	2054	.890	2478
.715	1719	.805	2076	.895	2504
.720	1737	.810	2097	.900	2530
.725	1755	.815	2119	.905	2556
.730	1774	.820	2142	.910	2583
.735	1793	.825	2164	.915	2610
.740	1811	.830	2187	.920	2637
.745	1831	.835	2210	.925	2665
* .750	1850	.840	2233	.930	2693
.755	1869	.845	2256	.935	2721
.760	1889	.850	2280	.940	2749
.765	1909	.855	2304	.945	2778
.770	1929	.860	2328	** .950	2807
.775	1949	.865	2352	.955	2836
.780	1970	.870	2377	.960	2866
.785	1991	.875	2402	.965	2895
				.970	2926
				.975	2956
				.980	2987
				.985	3018
				.990	3049
				.995	3081
				1.000	3113

* Specification requirements begin for this ordered thickness.

** Specification requirements end for this ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
0.940	1225	1.140	1618	1.340	2126
0.945	1234	1.145	1629	1.345	2141
* 0.951	1242	1.150	1640	1.350	2155
0.955	1251	1.155	1652	1.355	2170
0.960	1260	1.160	1663	1.360	2185
0.965	1269	1.165	1675	1.365	2199
0.970	1278	1.170	1686	1.370	2214
0.975	1287	1.175	1698	1.375	2229
0.980	1296	1.180	1709	1.380	2244
0.985	1305	1.185	1721	1.385	2260
0.990	1314	1.190	1733	1.390	2275
0.995	1323	1.195	1745	1.395	2290
1.000	1332	1.200	1757	1.400	2306
1.005	1342	1.205	1769	1.405	2321
1.010	1351	1.210	1781	1.410	2337
1.015	1361	1.215	1793	1.415	2353
1.020	1370	1.220	1806	1.420	2369
1.025	1380	1.225	1818	1.425	2385
1.030	1389	1.230	1831	1.430	2401
1.035	1399	1.235	1843	1.435	2417
1.040	1409	1.240	1856	1.440	2434
1.045	1419	1.245	1869	1.445	2450
1.050	1429	1.250	1881	1.450	2467
1.055	1439	1.255	1894	1.455	2483
1.060	1449	1.260	1907	1.460	2500
1.065	1459	1.265	1920	1.465	2517
1.070	1469	1.270	1933	1.470	2534
1.075	1479	1.275	1947	1.475	2551
1.080	1489	1.280	1960	1.480	2568
1.085	1500	1.285	1973	1.485	2585
1.090	1510	1.290	1987	1.490	2603
1.095	1521	1.295	2000	1.495	2620
1.100	1531	1.300	2014	** 1.500	2656
1.110	1552	1.310	2041	1.510	2673
1.115	1563	1.315	2055	1.515	2691
1.120	1574	1.320	2069	1.520	2709
1.125	1585	1.325	2083	1.525	2728
1.130	1596	1.330	2098	1.530	2746
1.135	1607	1.335	2112		

* Specification requirements begin for this ordered thickness.

** Specification requirements end for this ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
0.940	1997	1.140	2252	1.340	2481
0.945	2004	1.145	2258	1.345	2487
* 0.951	2010	1.150	2264	1.350	2492
0.955	2017	1.155	2270	1.355	2498
0.960	2024	1.160	2276	1.360	2503
0.965	2030	1.165	2282	1.365	2508
0.970	2037	1.170	2288	1.370	2514
0.975	2044	1.175	2294	1.375	2519
0.980	2050	1.180	2299	1.380	2525
0.985	2057	1.185	2305	1.385	2530
0.990	2063	1.190	2311	1.390	2535
0.995	2070	1.195	2317	1.395	2541
1.000	2077	1.200	2323	1.400	2546
1.005	2083	1.205	2329	1.405	2551
1.010	2090	1.210	2335	1.410	2557
1.015	2096	1.215	2340	1.415	2562
1.020	2102	1.220	2346	1.420	2567
1.025	2109	1.225	2352	1.425	2573
1.030	2115	1.230	2358	1.430	2578
1.035	2122	1.235	2363	1.435	2583
1.040	2128	1.240	2369	1.440	2588
1.045	2134	1.245	2375	1.445	2594
1.050	2141	1.250	2381	1.450	2599
1.055	2147	1.255	2386	1.455	2604
1.060	2153	1.260	2392	1.460	2609
1.065	2160	1.265	2398	1.465	2615
1.070	2166	1.270	2403	1.470	2620
1.075	2172	1.275	2409	1.475	2625
1.080	2178	1.280	2415	1.480	2630
1.085	2185	1.285	2420	1.485	2635
1.090	2191	1.290	2426	1.490	2640
1.095	2197	1.295	2431	1.495	2646
1.100	2203	1.300	2437	** 1.500	2651
1.105	2209	1.305	2443	1.505	2656
1.110	2215	1.310	2448	1.510	2661
1.115	2222	1.315	2454	1.515	2666
1.120	2228	1.320	2459	1.520	2671
1.125	2234	1.325	2465	1.525	2676
1.130	2240	1.330	2470	1.530	2681
1.135	2246	1.335	2476		

* Specification requirements begin for this ordered thickness.

** Specification requirements end for this ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
1.500	1955	1.695	2118	1.895	2274
* 1.501	1956	1.700	2122	1.900	2278
1.505	1959	1.705	2126	1.905	2281
1.510	1964	1.710	2130	1.910	2285
1.515	1968	1.715	2134	1.915	2289
1.520	1972	1.720	2138	1.920	2293
1.525	1976	1.725	2142	1.925	2296
1.530	1981	1.730	2146	1.930	2300
1.535	1985	1.735	2150	1.935	2304
1.540	1989	1.740	2154	1.940	2307
1.545	1994	1.745	2158	1.945	2311
1.550	1998	1.750	2162	1.950	2315
1.555	2002	1.755	2166	1.955	2319
1.560	2006	1.760	2170	1.960	2322
1.565	2011	1.765	2174	1.965	2326
1.570	2015	1.770	2178	1.970	2330
1.575	2019	1.775	2182	1.975	2333
1.580	2023	1.780	2186	1.980	2337
1.585	2028	1.785	2190	1.985	2341
1.590	2032	1.790	2193	1.990	2344
1.595	2036	1.795	2197	1.995	2348
1.600	2040	1.800	2201	2.000	2352
1.605	2044	1.805	2205	2.005	2355
1.610	2049	1.810	2209	2.010	2359
1.615	2053	1.815	2213	2.015	2363
1.620	2057	1.820	2217	2.020	2366
1.625	2061	1.825	2221	2.025	2370
1.630	2065	1.830	2224	2.030	2373
1.635	2069	1.835	2228	2.035	2377
1.640	2073	1.840	2232	2.040	2381
1.645	2077	1.845	2236	2.045	2384
1.650	2082	1.850	2240	2.050	2388
1.655	2086	1.855	2244	2.055	2391
1.660	2090	1.860	2247	2.060	2395
1.665	2094	1.865	2251	2.065	2398
1.670	2098	1.870	2255	2.070	2402
1.675	2102	1.875	2259	2.075	2406
1.680	2106	1.880	2263	2.080	2409
1.685	2110	1.885	2266	2.085	2413
1.690	2114	1.890	2270	2.090	2416

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
2.095	2420	2.295	2558	2.495	2689
2.100	2423	2.300	2561	2.500	2692
2.105	2427	2.305	2564	2.505	2695
2.110	2430	2.310	2568	2.510	2698
2.115	2434	2.315	2571	2.515	2701
2.120	2437	2.320	2574	2.520	2705
2.125	2441	2.325	2578	2.525	2708
2.130	2444	2.330	2581	2.530	2711
2.135	2448	2.335	2584	2.535	2714
2.140	2451	2.340	2588	2.540	2717
2.145	2455	2.345	2591	2.545	2720
2.150	2458	2.350	2594	2.550	2724
2.155	2462	2.355	2598	2.555	2727
2.160	2465	2.360	2601	2.560	2730
2.165	2469	2.365	2604	2.565	2733
2.170	2472	2.370	2608	2.570	2736
2.175	2476	2.375	2611	2.575	2739
2.180	2479	2.380	2614	2.580	2742
2.185	2483	2.385	2617	2.585	2746
2.190	2486	2.390	2621	2.590	2749
2.195	2490	2.395	2624	2.595	2752
2.200	2493	2.400	2627	2.600	2755
2.205	2497	2.405	2630	2.605	2758
2.210	2500	2.410	2634	2.610	2761
2.215	2503	2.415	2637	2.615	2764
2.220	2507	2.420	2640	2.620	2767
2.225	2510	2.425	2643	2.625	2770
2.230	2514	2.430	2647	2.630	2774
2.235	2517	2.435	2650	2.635	2777
2.240	2520	2.440	2653	2.640	2780
2.245	2524	2.445	2656	2.645	2783
2.250	2527	2.450	2660	2.650	2786
2.255	2531	2.455	2663	2.655	2789
2.260	2534	2.460	2666	2.660	2792
2.265	2537	2.465	2669	2.665	2795
2.270	2541	2.470	2673	2.670	2798
2.275	2544	2.475	2676	2.675	2801
2.280	2548	2.480	2679	2.680	2804
2.285	2551	2.485	2682	2.685	2807
2.290	2554	2.490	2685	2.690	2811

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
2.695	2814	2.810	2883	2.925	2951
2.700	2817	2.815	2886	2.930	2954
2.705	2820	2.820	2889	2.935	2957
2.710	2823	2.825	2892	2.940	2960
2.715	2826	2.830	2895	2.945	2963
2.720	2829	2.835	2898	2.950	2965
2.725	2832	2.840	2901	2.955	2968
2.730	2835	2.845	2904	2.960	2971
2.735	2838	2.850	2907	2.965	2974
2.740	2841	2.855	2910	2.970	2977
2.745	2844	2.860	2913	2.975	2980
2.750	2847	2.865	2916	2.980	2983
2.755	2850	2.870	2919	2.985	2986
2.760	2853	2.875	2922	2.990	2989
2.765	2856	2.880	2925	2.995	2991
2.770	2859	2.885	2927	** 3.000	2994
2.775	2862	2.890	2930	3.005	2997
2.780	2865	2.895	2933	3.010	3000
2.785	2868	2.900	2936	3.015	3003
2.790	2871	2.905	2939	3.020	3006
2.795	2874	2.910	2942	3.025	3009
2.800	2877	2.915	2945	3.030	3012
2.805	2880	2.920	2948		

* Specification requirements begin for this ordered thickness.

** Specification requirements end for this ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-VI. Minimum required ballistic limits - caliber 14.5 mm API, BS41 projectiles at 0° obliquity.

	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
	2.950	2615	3.135	2739	3.320	2857
	2.955	2619	3.140	2742	3.325	2860
	2.960	2622	3.145	2746	3.330	2864
	2.965	2626	3.150	2749	3.335	2867
	2.970	2629	3.155	2752	3.340	2870
	2.975	2633	3.160	2755	3.345	2873
	2.980	2636	3.165	2759	3.350	2876
	2.985	2640	3.170	2762	3.355	2879
	2.990	2643	3.175	2765	3.360	2882
	2.995	2647	3.180	2769	3.365	2885
*	3.001	2650	3.185	2772	3.370	2888
	3.005	2653	3.190	2775	3.375	289l
	3.010	2656	3.195	2779	3.380	2895
	3.015	2660	3.200	2782	3.385	2898
	3.020	2663	3.205	2785	3.390	290l
	3.025	2666	3.210	2788	3.395	2904
	3.030	2670	3.215	279l	3.400	2907
	3.035	2673	3.220	2795	3.405	29l0
	3.040	2676	3.225	2798	3.410	29l3
	3.045	2680	3.230	280l	3.415	29l6
	3.050	2683	3.235	2804	3.420	2920
	3.055	2686	3.240	2808	3.425	2923
	3.060	2689	3.245	28ll	3.430	2926
	3.065	2693	3.250	28l4	3.435	2929
	3.070	2696	3.255	28l7	3.440	2932
	3.075	2699	3.260	2820	3.445	2935
	3.080	2703	3.265	2823	3.450	2938
	3.085	2706	3.270	2826	3.455	2941
	3.090	2709	3.275	2829	3.460	2944
	3.095	2713	3.280	2833	3.465	2948
	3.100	2716	3.285	2836	3.470	2950
	3.105	2719	3.290	2839	3.475	2953
	3.110	2722	3.295	2842	3.480	2956
	3.115	2726	3.300	2845	3.485	2959
	3.120	2729	3.305	2848	3.490	2962
	3.125	2732	3.310	2851	3.495	2965
	3.130	2736	3.315	2854	3.500	2968

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-VI. Minimum required ballistic limits - caliber 14.5 mm API, BS41 projectiles at 0° obliquity - Continued.

Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps	Thickness, in.	Required BL(P), fps
3.505	297l	3.690	308l	3.875	3185
3.510	2974	3.695	3084	3.880	3187
3.5l5	2978	3.700	3087	3.885	3l90
3.520	2980	3.705	3090	3.890	3l93
3.525	2983	3.710	3093	3.895	3l95
3.530	2986	3.715	3096	3.900	3l98
3.535	2989	3.720	3099	3.905	3200
3.540	2992	3.725	3l02	3.910	3203
3.545	2995	3.730	3104	3.915	3206
3.550	2998	3.735	3107	3.920	3208
3.555	300l	3.740	31l0	3.925	32ll
3.560	3004	3.745	31l3	3.930	32l4
3.565	3007	3.750	31l6	3.935	32l6
3.570	30l0	3.755	31l9	3.940	32l9
3.575	30l3	3.760	3122	3.945	3222
3.580	3016	3.765	3125	3.950	3224
3.585	30l9	3.770	3127	3.955	3226
3.590	3022	3.775	3130	3.960	3229
3.595	3025	3.780	3133	3.965	3232
3.600	3028	3.785	3136	3.970	3234
3.605	3031	3.790	3138	3.975	3237
3.610	3034	3.795	3141	3.980	3240
3.615	3037	3.800	3144	3.985	3242
3.620	3040	3.805	3147	3.990	3245
3.625	3043	3.810	3150	3.995	3248
3.630	3046	3.815	3152	** 4.000	3250
3.635	3049	3.820	3155	4.005	3252
3.640	3052	3.825	3158	4.010	3255
3.645	3055	3.830	3160	4.015	3257
3.650	3058	3.835	3163	4.020	3260
3.655	3061	3.840	3166	4.025	3262
3.660	3064	3.845	3168	4.030	3265
3.665	3067	3.850	3l7l	4.035	3267
3.670	3070	3.855	3l74	4.040	3270
3.675	3073	3.860	3l77	4.045	3272
3.680	3075	3.865	3l79	4.050	3275
3.685	3078	3.870	3l82		

* Specification requirements begin for this test at ordered thickness.

** Specification requirements end for this test at ordered thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - AT, TE
Navy - SH
Air Force - 84, 99
DLA - IS

Preparing activity
Army - MR

(Project 9535-0624)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-46063H (MR)	**2. DOCUMENT DATE (YYMMDD)** 980914

DOCUMENT TITLE ARMOR PLATE , ALUMINUM ALLOY, 7039

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Research Laboratory Weapons and Material Research Dirctorate	b. TELEPHONE *Include Area Code)* (1) Commercial (410)306-0725 (2) AUTOVON 458-0725
c. ADDRESS *(Include Zip Code)Commander* ARL/WMRD ATTN; AMSRL/WM-M Aberdeen Proving Ground, MD 21005-5069	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.